

ACM CCS 2025 KEYNOTE PRESENTATION

Title: Autonomous Vulnerability Analysis, Triaging, and Repair: A Historical Perspective

Speaker: Giovanni Vigna

Affiliation: University of California, Santa Barbara (UCSB) & SSLab

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Venue: ACM Conference on Computer and Communications Security (CCS 2025), Taipei, Taiwan

URL: <https://sites.cs.ucsb.edu/~vigna/>

ABOUT THE SPEAKER

Giovanni Vigna is a Professor of Computer Science at the University of California, Santa Barbara, and co-director of the SecLab. He is also the co-founder of Wombat Security Technologies. His research interests include malware analysis, vulnerability assessment, web security, and the application of machine learning to security problems. He has co-authored hundreds of publications and has been a key organizer of CTF competitions and DARPA Cyber challenges (CGC, AIXCC).

KEYNOTE SUMMARY

In this keynote, Professor Vigna explores the evolution of automated security analysis over the past decade, tracing the line of development from the 2016 DARPA Cyber Grand Challenge (CGC) to the recent 2024–2025 DARPA AI Cyber Challenge (AIXCC).

Core Themes Covered:

1. The Evolution of Automation: Traces how the security community moved from purely static analysis and symbolic execution (CGC era) to hybrid architectures combining deep semantic models (LLMs) with program analysis tools.
2. Execution-Aware Reasoning: Highlights the critical limitation of using LLMs in isolation for vulnerability proof or exploit generation. Real-world systems require grounding LLMs in concrete program execution states (interactive feedback loops with gdb, sanitisers, and test suites).
3. The AI Cyber Challenge (AIXCC): Explores the architecture and design of modern Cyber Reasoning Systems (CRSs) such as ATLANTIS and OSS-CRS, focusing on autonomous patch generation, harness generation, and automated verification.
4. The Road Ahead: Identifies the remaining open gaps in fully autonomous vulnerability lifecycles, emphasizing safety, rate-limits, context-window management, and the challenge of high-fidelity program emulation.

ALIGNMENT WITH CMATRIX

Vigna's keynote directly informs the CMatrix multi-agent architecture:

- Re-enforces the necessity of integrating dynamic feedback (execution-state observation) into the LLM planning cycle rather than relying on static text prompts.
- Identifies the core tasks (reconnaissance, vulnerability identification, exploitation validation, and reporting) that must be partitioned among specialized agent roles.